

UNIT 2

LIFE CYCLES OF A BUTTERFLY AND A FROG

Learning objectives:

- Define a life cycle and explain the concept of metamorphosis with examples from butterflies and frogs.
- Describe the stages in the life cycle of a butterfly, including egg, larva (caterpillar), pupa (chrysalis), and adult.
- Explain the stages in the life cycle of a frog, including egg, tadpole, tadpole with legs, and adult frog.
- Compare the life cycles of a butterfly and a frog in terms of their stages and movement of young forms.

LIFE CYCLES

Every living thing goes through stages as it grows. This journey from birth to adulthood is called a **life cycle**. In this chapter, we will explore the life cycles of a **butterfly** and a **frog**, which go through big changes called **metamorphosis**.

LIFE CYCLE OF A BUTTERFLY

Butterflies are colourful insects that go through four stages:

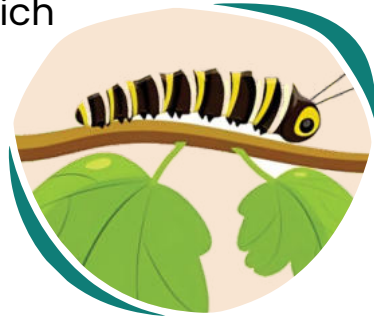
1. Egg Stage

- Life begins as a tiny egg laid on leaves.
- These eggs are small and can be round, oval, or cylindrical.



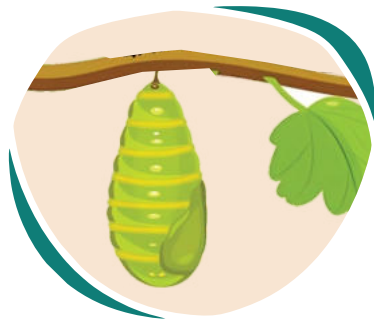
2. Larva Stage (Caterpillar)

- The egg hatches into a caterpillar, which eats leaves to grow.
- As it grows, the caterpillar sheds its skin multiple times.



3. Pupa Stage (Chrysalis)

- The caterpillar wraps itself in a protective shell called a chrysalis.
- Inside the chrysalis, the caterpillar transforms into a butterfly.



4. Adult Stage (Butterfly)

- A butterfly emerges, dries its wings, and flies away.
- It finds flowers and lays eggs to start the cycle again.

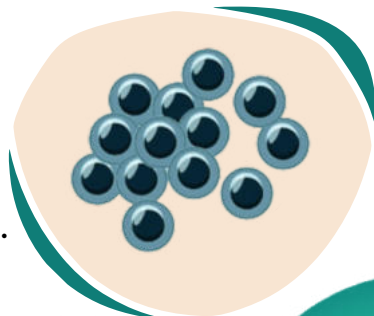


LIFE CYCLE OF A FROG

Frogs are amphibians, meaning they live in water and on land. Their life cycle also has four stages:

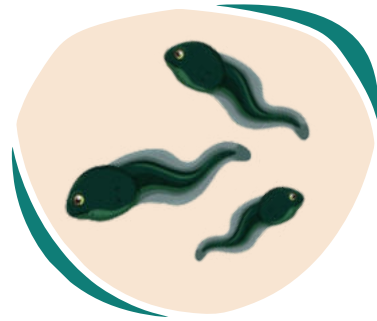
1. Egg Stage

- Frogs lay eggs in clusters called frogspawn in water.
- Frogspawn looks like tiny jelly bubbles.



2. Tadpole Stage

- The eggs hatch into tadpoles that swim in water, breathe through gills, and eat plants.



3. Tadpole with Legs

- Tadpoles grow legs and lose their tails as they develop.
- At this stage, they begin to live on both land and water.



4. Adult Frog

- The tadpole fully transforms into an adult frog with lungs for breathing air and strong legs for jumping.



COMPARISON BETWEEN BUTTERFLY AND FROG LIFE CYCLES

Feature	Butterfly	Frog
Eggs Laid On	Leaves	Water
Movement of Young	Caterpillar crawls	Tadpole swims
Stages	Egg, Larva, Pupa, Adult	Egg, Tadpole, Froglet, Adult

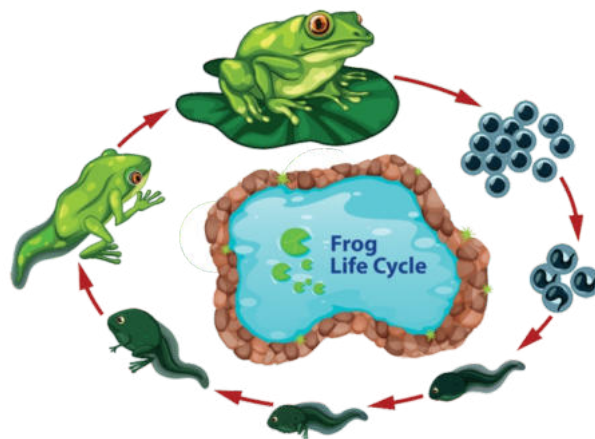
Fun Fact

- A butterfly's wings are covered with tiny scales that reflect light.
- Frogs can jump up to 20 times their body length!
- Some frogs live for more than 10 years, while some butterflies live for only a week.



Fill in the blanks.

1. A butterfly's life starts as an _____.
2. A _____ is the stage where a caterpillar transforms into a butterfly.
3. Frog eggs are laid in clusters called _____.
4. Tadpoles breathe through _____ while living in water.
5. Adult frogs return to _____ to lay eggs.
6. A chrysalis is a protective shell formed by a _____.



Write "T" for True or "F" for False.

1. A caterpillar is the adult stage of a butterfly.

2. Frogspawn looks like jelly bubbles.

3. Tadpoles can breathe on land.

4. The pupa stage of a butterfly is also called a chrysalis.

5. Adult frogs breathe with their tails.

6. Frogs and butterflies do not go through metamorphosis.





Choose the correct answer.

1. What is the first stage in the life cycle of butterfly?

- Pupa
 - Larva
 - Egg
-

2. During which stage does a caterpillar transform into a butterfly?

- Egg
 - Pupa
 - Adult
-

3. What is the purpose of gills in fish?

- Breathing
 - Eating
 - To swim
-

4. Where do frogs lay their eggs in?

- Trees
 - Water
 - Leaves
-

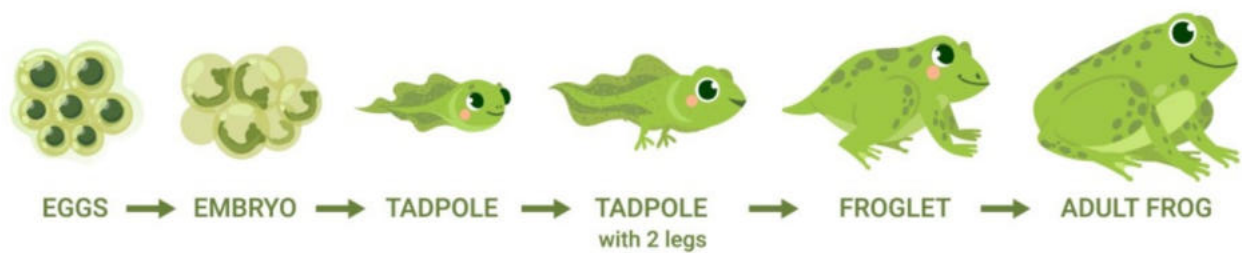


5. Which is the adult stage of a caterpillar?

- Pupa
- Caterpillar
- Butterfly

6. What helps tadpoles survive in water?

- Gills
- Wings
- Strong legs



Answer the following questions.

Q1: What are the four stages of a butterfly's life cycle?

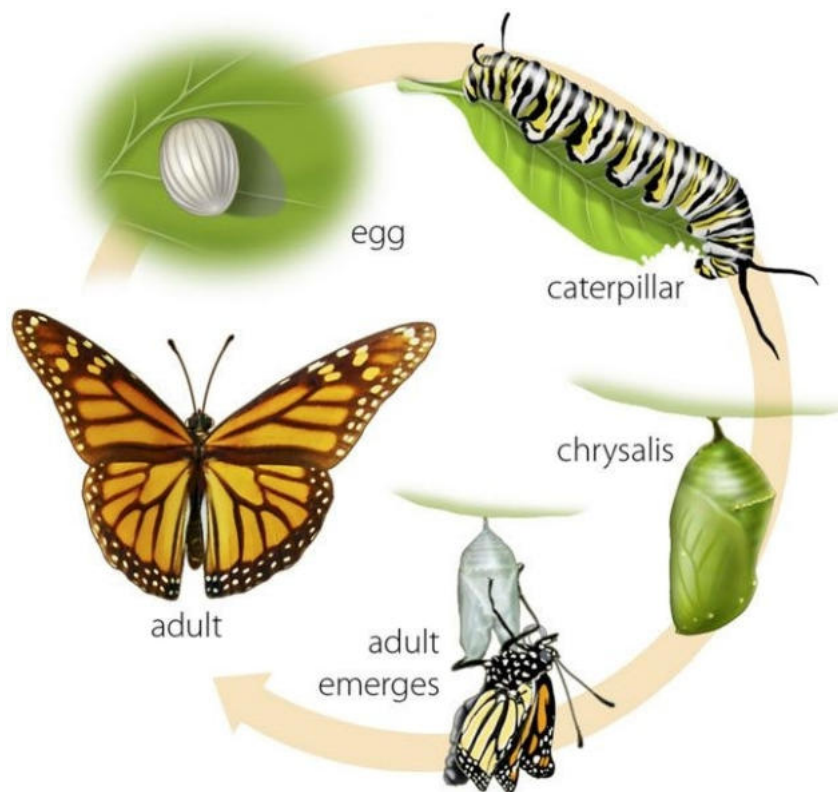
Q2: How does a tadpole survive in water?

Q3: What is the difference between frogspawn and butterfly eggs?

Q4: What happens inside a chrysalis?

Q5: Why do adult frogs need strong legs?

Q6: Compare the movements of a caterpillar and a tadpole.





Create a Life Cycle Chart



ACTIVITY

- Draw the life cycle of either a butterfly or a frog in your notebook.
- Label each stage and use different colours for each stage.

